

Sea_Level_Rise_Timeseries

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Introduction

Sea level acts as an important indicator of long-term environmental and climatic change events. This study analyzes Mumbai's monthly sea-level data from 1878–1961 using a continuous gap-free dataset containing 1,008 observations extracted from National oceanography center (<https://psmsl.org/data/obtaining/stations/43.php>). The objective of the study is to examine long-term sea-level trends, seasonal variability, and forecasting behavior using time-series analysis techniques.

The analysis includes data visualization, monthly and annual mean analysis, STL decomposition, trend estimation, Mann–Kendall and Theil–Sen statistical tests, and ARIMA forecasting. Various graphical methods such as time-series plots, anomaly stripes, heatmaps, and ridge plots were used to understand temporal and seasonal patterns in sea level rise.

The study helps identify historical sea-level rise patterns and evaluates the strengths and limitations of statistical forecasting models for environmental time-series data.

Objectives

1. To analyze long-term sea-level variations from 1878–1961 using monthly observational data.
2. To identify seasonal and temporal patterns in sea-level fluctuations using time-series visualization techniques.
3. To estimate and evaluate long-term sea-level trends using Linear Regression, Mann–Kendall, and Theil–Sen methods.
4. To decompose the time series into trend, seasonal, and residual components using STL decomposition.
5. To develop and evaluate an ARIMA forecasting model for short-term sea-level prediction.

Load Required Libraries:

```
library(tidyverse) #used for data manipulation and visualization
library(forecast)  #used for time-series forecasting and ARIMA modeling
library(trend)     #used for Mann-Kendall and Theil-Sen trend analysis
library(zoo)       #used for working with regular and irregular time series data
library(RColorBrewer) #used for creating color palettes for visualizations
library(viridis)   #used for creating color palettes for visualizations
library(ggribbles) #used for creating ridge plots for visualizing distributions
```

Data Loading and Cleaning

The original sea level dataset contains monthly sea level observations for Mumbai recorded in decimal-year format. The raw data was cleaned and then transformed into a structured monthly time-series data format suitable for timeseries analysis.

Only observations with: - valid sea level measurements (No NA's in data, as for forecasting we need continue data without any gaps), - quality flag = 0, - and continuous monthly records

A strict continuous segment from 1878–1961 was selected because time-series forecasting and decomposition methods require uninterrupted observations. Post-1961 records contain multiple fragmented gaps, including long missing periods during 1995–2004 and several additional discontinuities after 2012.

```
dat <- read.table("D:\\R_JP\\Assi\\43.rlrdata.txt",
                 sep = ";",
                 header = FALSE,
                 col.names = c("dec_year", "sl_mm", "flag", "flag2"),
                 #dce_year = Decimal year (e.g., 1878.0833 for February 1878)
                 #sl_mm = Sea level in millimeters
                 #Extra columns (flag, flag2) for quality control
                 na.strings = "-99999",
                 )
```

```
dat$year <- floor(dat$dec_year)
#floor removes the decimal part, giving the integer year
dat$month <- floor((dat$dec_year - dat$year) * 12) + 1
# +1 is for the month configuration
#formula year-year*12+1 gives the month number (1-12) based on the
#decimal part of the year.
dat$date <- as.Date(paste(dat$year, dat$month, "15", sep = "-"))
#this takes 15th day as a date as the dataset
```

```
seg <- dat %>%
  filter(!is.na(sl_mm), flag == 0,
         year >= 1878, year <= 1961) %>%
  arrange(date)
```

```
#creates a time series object with monthly frequency.
sl_ts <- ts(seg$sl_mm, frequency = 12, start = c(1878, 1))
```

Annual Mean Calculation:

Monthly sea level observations were aggregated into annual mean values to reduce short-term seasonal variability and emphasize long-term temporal trends. Annual averaging provides a clearer representation of gradual sea level change and is commonly used in climatological and oceanographic trend studies.

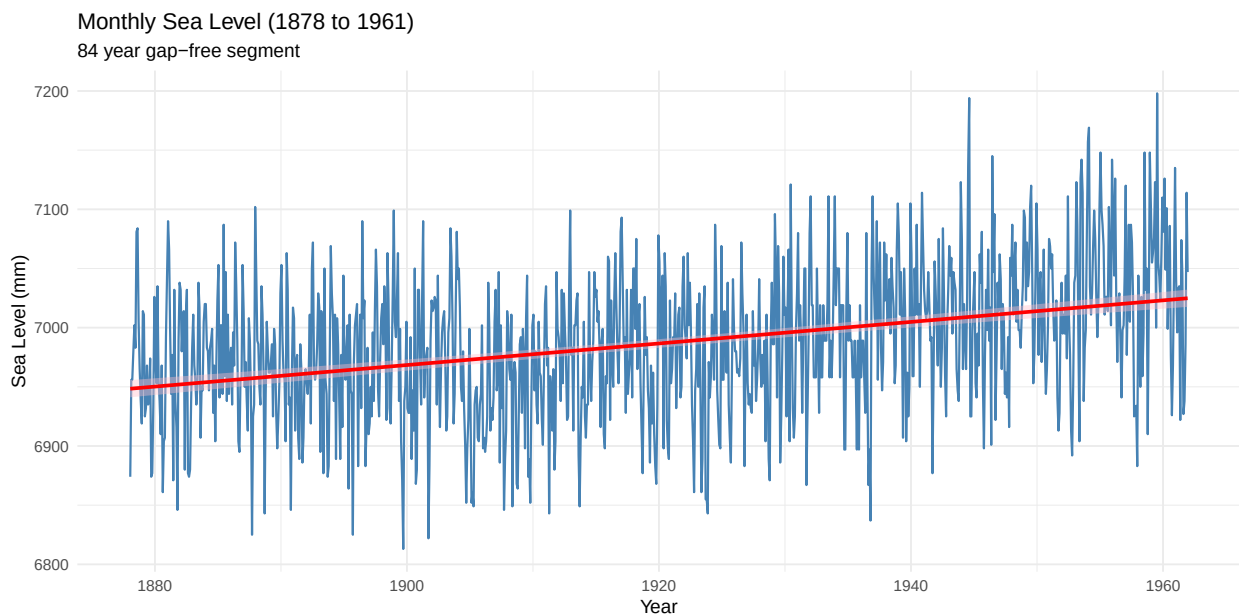
```
annual <- seg %>%
  group_by(year) %>%
  #calculate the mean sea level for each year, ignoring NA values
  summarise(mean_sl = mean(sl_mm), .groups = "drop")
```

```
#ungroup the data after summarization
```

Monthly Sea Level Visualization:

The monthly sea level time series was visualized to examine short-term variability, long-term tendencies, and overall continuity within the selected 84 year data segment. A linear regression smoother was added to identify the general long-term trend in the monthly observations from 1878 to 1961.

```
p1 <- ggplot(seg, aes(x = date, y = sl_mm)) +  
  geom_line(col = "steelblue", linewidth = 0.6) +  
  #adds a line plot to show the monthly sea level variations  
  geom_smooth(method = "lm", col = "red", se = TRUE, fill = "pink") +  
  #adds a linear regression line with confidence interval  
  labs(title = "Monthly Sea Level (1878 to 1961)",  
        subtitle = "84 year gap-free segment",  
        x = "Year", y = "Sea Level (mm)") +  
  theme_minimal()  
print(p1)
```

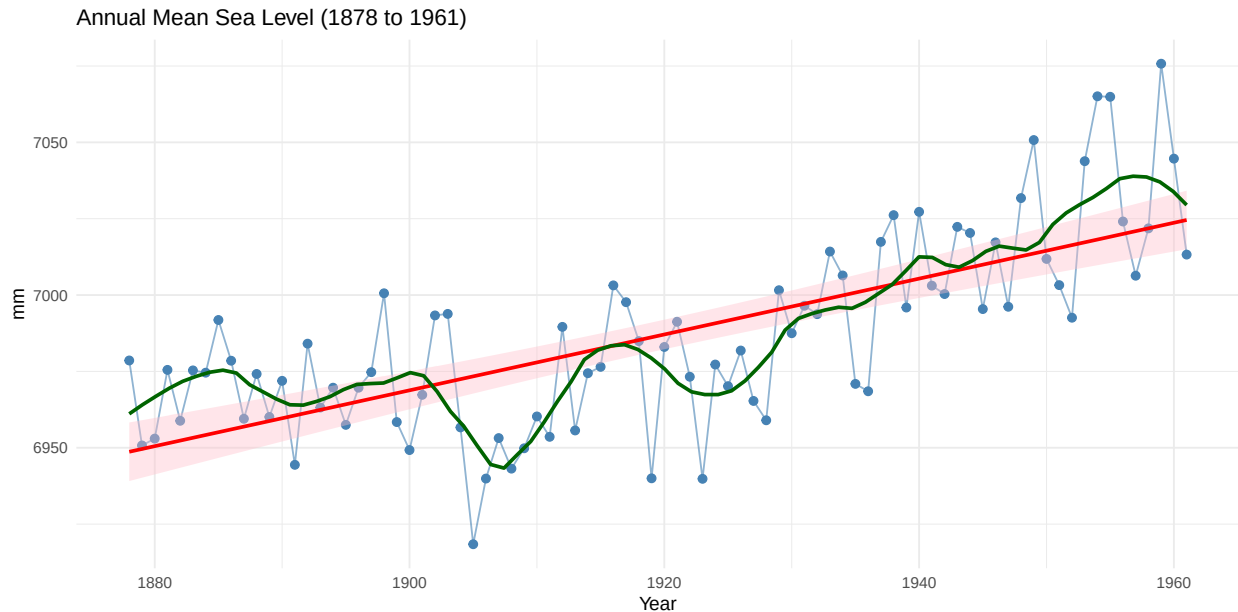


Annual Mean Sea Level Trend

Annual mean sea level values were plotted to investigate long-term variability and possible acceleration patterns over time. Both linear regression and LOESS smoothing techniques were applied: 1. Linear regression estimates the overall monotonic trend. 2. LOESS smoothing highlights localized nonlinear fluctuations.

```
p2 <- ggplot(annual, aes(x = year, y = mean_sl)) +  
  geom_point(col = "steelblue", size = 2) +  
  #adds points to show the annual mean sea level values  
  geom_line(col = "steelblue", alpha = 0.6) +
```

```
#adds a line plot to connect the annual mean sea level values
geom_smooth(method = "lm", col = "red", fill = "pink") +
#adds a linear regression line with confidence interval to show the overall trend
geom_smooth(method = "loess", span = 0.2, col = "darkgreen", se = FALSE) +
#adds a LOESS smoothing line to show localized nonlinear fluctuations
labs(title = "Annual Mean Sea Level (1878 to 1961)",
      x = "Year", y = "mm") +
theme_minimal()
print(p2)
```



Sea Level Anomaly Stripes

Sea level anomaly stripes were generated relative to the segment mean baseline. This visualization emphasizes periods of above-average and below-average sea level conditions using a continuous color gradient. Climate-stripe style visualizations provide an intuitive representation of long-term environmental change.

```
baseline <- mean(annual$mean_sl) #Calculates the average sea level across the
#entire study period of 84 years for baseline.
annual$anomaly <- annual$mean_sl - baseline #calculating anomaly

p3 <- ggplot(annual, aes(x = year, y = 1, fill = anomaly)) +
#Using annual data for visualization
geom_col(width = 1, show.legend = FALSE) +
scale_x_continuous(expand = c(0, 0), breaks = seq(1880, 1960, 20)) +
#setting x-axis breaks every 20 years
scale_y_continuous(expand = c(0, 0)) +
scale_fill_gradientn(colors = rev(brewer.pal(11, "RdBu")),
                    limits = c(-max(abs(annual$anomaly)),
                               max(abs(annual$anomaly)))) +
#setting the limits of the color scale to be symmetric around
#zero based on the maximum absolute anomaly value
```

```

labs(title = "Sea Level Anomaly Stripes (1878 to 1961)",
     subtitle = sprintf("Baseline = segment mean (%.0f mm)", baseline),
     x = NULL, y = NULL) +
theme_minimal() +
theme(axis.text.y = element_blank(), axis.ticks.y = element_blank(),
      panel.grid = element_blank())
#removing y-axis text, ticks, and grid for a cleaner stripe visualization

print(p3)

```

Sea Level Anomaly Stripes (1878 to 1961)
Baseline = segment mean (6987 mm)

Calendar Heatmap

A calendar heatmap was produced to visualize monthly sea level variability across multiple decades simultaneously.

This approach helps identify: 1. seasonal patterns, 2. recurring anomalies, 3. and interannual variability in sea level behavior.

```

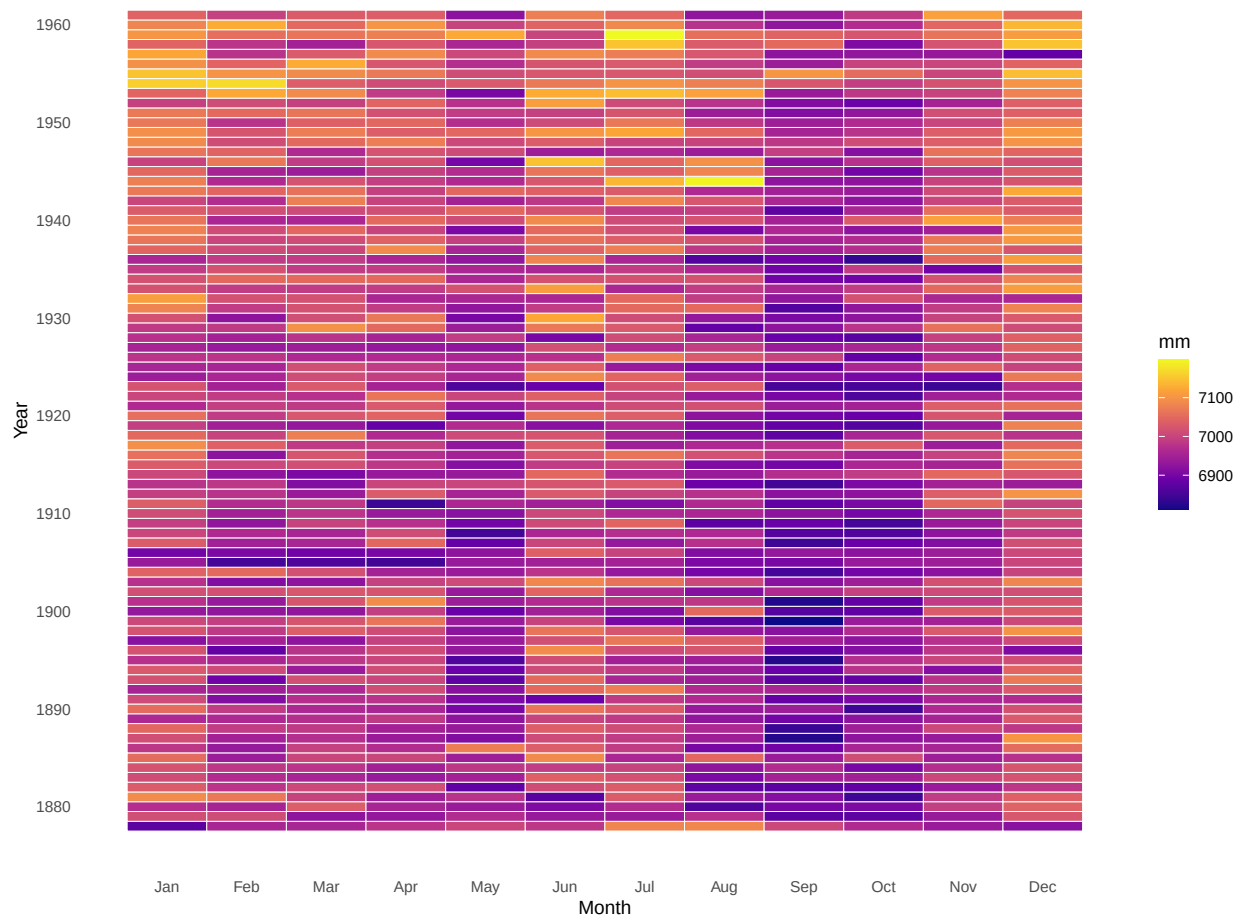
cal <- seg %>% #group the data by year and month
  group_by(year, month) %>%
  #calculate the average sea level for each month of each year, ignoring NA values
  summarise(avg_sl = mean(sl_mm), .groups = "drop")

p4 <- ggplot(cal, aes(x = month, y = year, fill = avg_sl)) +
  geom_tile(color = "white", linewidth = 0.1) +
  #adds tiles to create the heatmap with white borders between tiles
  scale_fill_viridis_c(option = "plasma", name = "mm") +
  #using the plasma color palette from viridis for the fill colors
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  scale_y_continuous(breaks = seq(1880, 1960, 10)) +
  labs(title = "Sea Level Calendar Heatmap (1878 to 1961)",
       x = "Month", y = "Year") +
  theme_minimal() +
  theme(panel.grid = element_blank())
#removing grid lines for a cleaner heatmap visualization

print(p4)

```

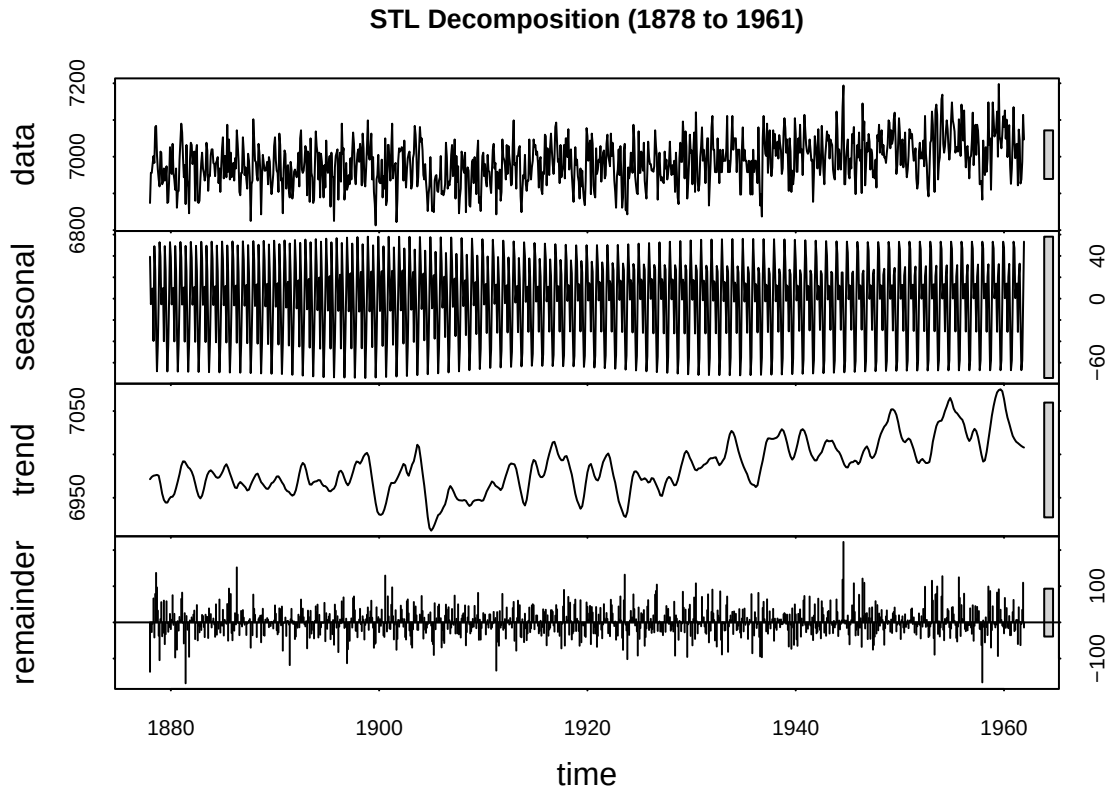
Sea Level Calendar Heatmap (1878 to 1961)



STL Decomposition

Density ridge plots were used to compare the statistical distribution of sea level observations between decades. This visualization highlights gradual shifts in the central tendency and spread of sea level distributions through time.

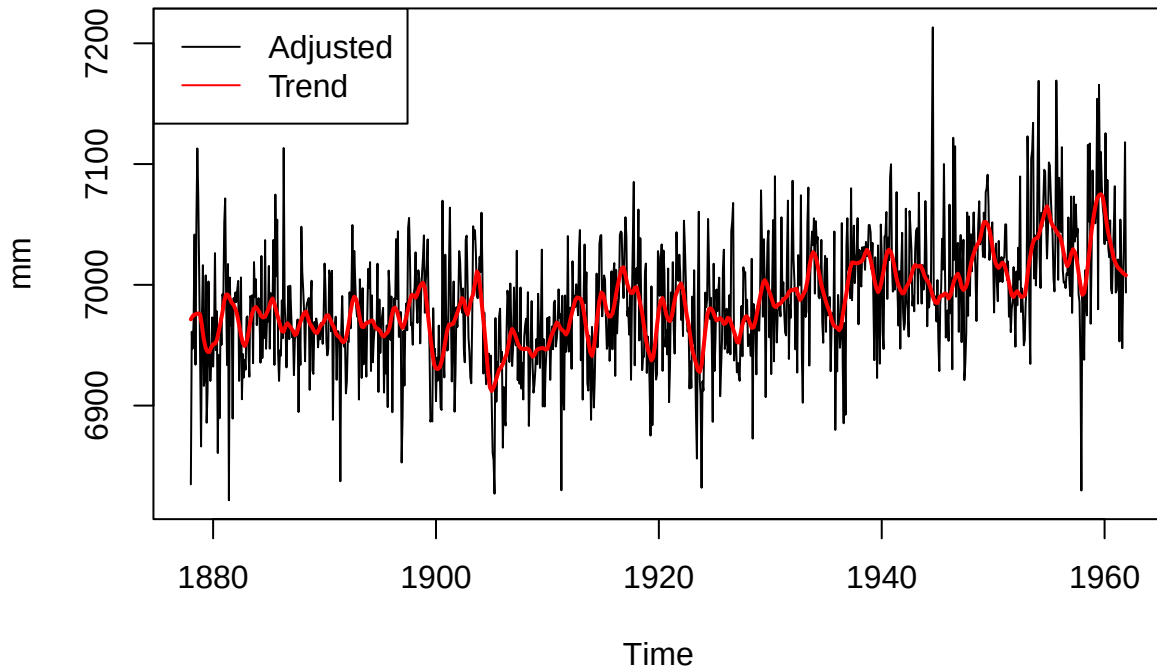
```
stl_fit <- stl(sl_ts, s.window = 35, t.window = 19, robust = TRUE)
#STL decomposition with specified seasonal and trend window sizes,
#and robust fitting to reduce the influence of outliers
plot(stl_fit, main = "STL Decomposition (1878 to 1961)")
```



```
sa <- sl_ts - stl_fit$time.series[, "seasonal"]
#calculating the seasonally adjusted series by raw timeseries
plot(sa, main = "Seasonally Adjusted (1878 to 1961)", ylab = "mm")
#plots the seasonally adjusted time series

lines(stl_fit$time.series[, "trend"], col = "red", lwd = 2)
legend("topleft", legend = c("Adjusted", "Trend"), col = c("black", "red"), lty = 1)
```

Seasonally Adjusted (1878 to 1961)



Trend Analysis

Multiple statistical approaches were used to evaluate long-term sea level trends and improve robustness of interpretation.

The analysis included: 1. Ordinary Least Squares (OLS) regression, 2. Mann–Kendall trend test, 3. Theil–Sen slope estimation, 4. and Seasonal Mann–Kendall testing.

These methods collectively assess both linear and non-parametric trend behavior while accounting for seasonal structure in the dataset.

```
lm_fit <- lm(mean_sl ~ year, data = annual)
#fits a linear regression model to the annual mean sea level data
print(summary(lm_fit)$coefficients)

##              Estimate Std. Error  t value    Pr(>|t|)
## (Intercept) 5231.5268264 192.5275568 27.172873 9.079882e-43
## year         0.9143574   0.1002929  9.116872 4.213696e-14

#prints the coefficients of the linear regression model,
#including the intercept and slope, along with their standard errors, t-values,
#and p-values to assess the statistical significance of the trend

cat(sprintf("Trend: %.3f mm/year, R² = %.3f\n\n",
            coef(lm_fit)[2], summary(lm_fit)$r.squared))
```



```

## Trend: 0.914 mm/year,  $R^2 = 0.503$ 

#prints the estimated trend in millimeters per year

cat("-----\n\n")

## -----

mk <- mk.test(annual$mean_sl)
#performs the Mann-Kendall trend test on the annual mean sea level data
print(mk)

##
## Mann-Kendall trend test
##
## data: annual$mean_sl
## z = 6.6832, n = 84, p-value = 2.338e-11
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##          S          varS          tau
## 1.731000e+03 6.700767e+04 4.966289e-01

df_sen <- data.frame(sl = annual$mean_sl, yr = annual$year)
#creates a data frame with the annual mean sea level (sl)
#and corresponding years (yr) to be used for Theil-Sen slope estimation
sen <- zyp::zyp.sen(sl ~ yr, data = df_sen)
#calculates the Theil-Sen slope estimator for the annual mean sea level data
cat(sprintf("\nTHEIL-SEN: %.3f mm/year\n\n", coef(sen)[2]))

##
## THEIL-SEN: 0.862 mm/year

#prints the estimated Theil-Sen slope in millimeters per year

smk <- smk.test(sl_ts)
#performs the Seasonal Mann-Kendall test on the monthly sea level time series
print(smk)

##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: sl_ts
## z = 13.405, p-value < 2.2e-16
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##          S          varS
## 12015 803259

```

Forecasting Analysis

Timeseries forecasting was performed using seasonal ARIMA modeling to evaluate short-term predictive performance.

The dataset was divided into: 1. training period (1878–1959) 2. testing period (1960–1961)

Forecast skill was evaluated using standard performance metrics including: 1. Mean Absolute Percentage Error (MAPE), 2. prediction bias, 3. and Theil's U statistic.

This analysis assesses the ability of historical sea level dynamics to predict future observations within the continuous segment.

```
train <- window(sl_ts, end = c(1959, 12))
#creates a training time series object that includes all monthly sea level data
test  <- window(sl_ts, start = c(1960, 1))
#creates a test time series object that includes the monthly sea level data
cat(sprintf("Training: %d months\n", length(train)))

## Training: 984 months

#prints the number of months in the training dataset
cat(sprintf("Test: %d months\n\n", length(test)))

## Test: 24 months

#prints the number of months in the test dataset

fit <- auto.arima(train, seasonal = TRUE, stepwise = FALSE)
#fits an ARIMA model to the training time series data using the auto.arima function
#(stepwise = FALSE) to find the optimal model parameters

cat("ARIMA model:\n")

## ARIMA model:
print(summary(fit))

## Series: train
## ARIMA(0,1,3)(2,0,0)[12]
##
## Coefficients:
##          ma1      ma2      ma3      sar1      sar2
##      -0.7309 -0.1624 -0.0916  0.3012  0.1837
## s.e.   0.0321  0.0409  0.0314  0.0318  0.0325
##
## sigma^2 = 2628: log likelihood = -5264.78
## AIC=10541.55  AICc=10541.64  BIC=10570.9
##
## Training set error measures:
##              ME      RMSE      MAE      MPE      MAPE
## Training set 2.549222 51.10732 40.51476 0.03113943 0.5796608
##              MASE      ACF1
## Training set 0.7916921 0.00153731

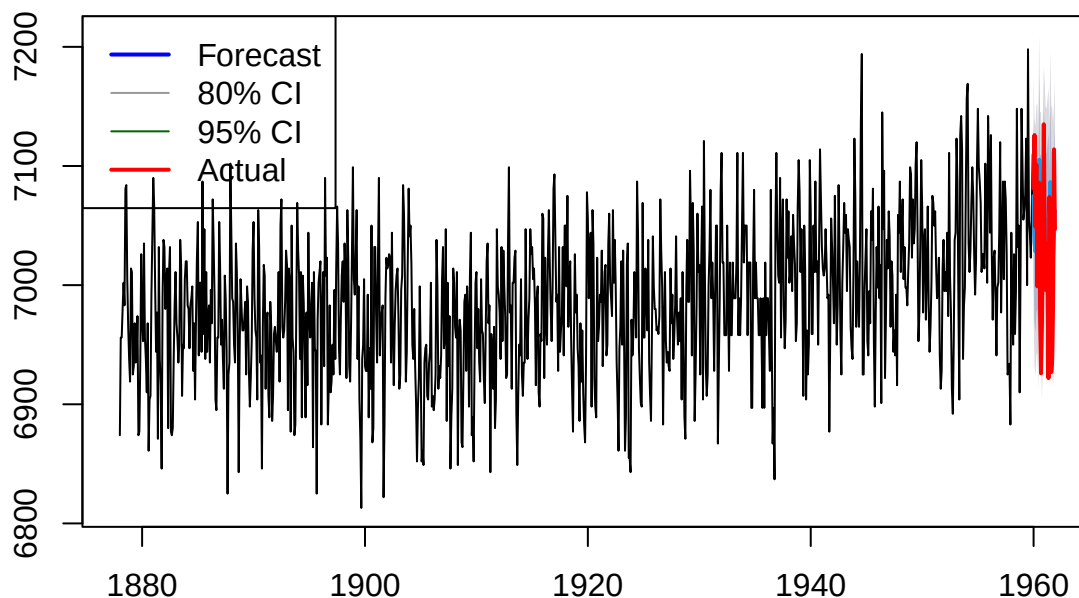
fc <- forecast(fit, h = 24)
#generates forecasts for the next 24 months (2 years) using the fitted ARIMA model
plot(fc, main = "ARIMA Forecast vs Actual (Jan 1960 to Dec 1961)")
```

```

lines(test, col = "red", lwd = 2)
legend("topleft", legend = c("Forecast", "80% CI", "95% CI", "Actual"),
      col = c("blue", "grey60", "darkgreen", "red"), lty = 1, lwd = c(2,1,1,2))

```

ARIMA Forecast vs Actual (Jan 1960 to Dec 1961)



```

#adds a legend to the plot to differentiate between the forecasted values (blue),
#the 80% confidence interval (grey60), the 95% confidence interval (darkgreen),
#and the actual observed values (red)

```

```

acc <- accuracy(fc, test)
#calculates the accuracy of the ARIMA forecasts
print(acc)

```

```

##              ME      RMSE      MAE      MPE      MAPE
## Training set  2.549222 51.10732 40.51476  0.03113943 0.5796608
## Test set     -17.904072 61.77125 49.65564 -0.26176831 0.7087946
##              MASE      ACF1 Theil's U
## Training set  0.7916921 0.00153731      NA
## Test set     0.9703125 0.16235458 0.8053103

```

```

cat(sprintf("\nTheil's U: %.2f (%s)\n", acc[2, "Theil's U"],
  ifelse(acc[2, "Theil's U"] < 1, "beats naive", "worse than naive")))

```

```

##

```

```
## Theil's U: 0.81 (beats naive)
```

```
#prints Theil's U statistic
```

Residual Diagnostics

Residual diagnostic tests were conducted to evaluate the statistical adequacy of the decomposition and forecasting models.

Autocorrelation analysis, normality testing, and Ljung–Box statistics were used to determine whether residual components behaved as approximately random noise.

These diagnostics help assess model reliability and identify remaining structure not captured by the fitted models.

```
resid <- stl_fit$time.series[, "remainder"]
```

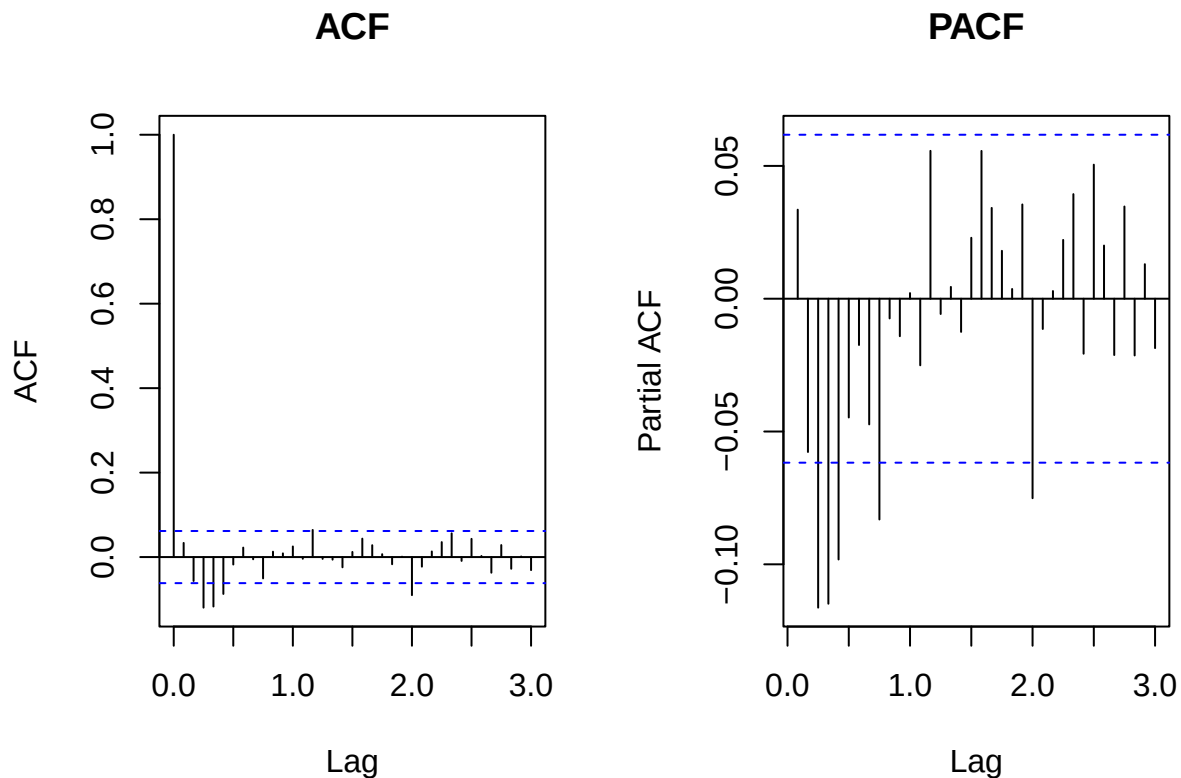
```
#extracts the remainder (residual) component from the STL decomposition
```

```
par(mfrow = c(1,2))
```

```
acf(resid, lag.max = 36, main = "ACF")
```

```
#plots the autocorrelation function (ACF) of the residuals up to a lag of 36 months
```

```
pacf(resid, lag.max = 36, main = "PACF")
```



```
#plots the partial autocorrelation function (PACF) of the residuals up to a  
#lag of 36 months
```

```

par(mfrow = c(1,1))

lb <- Box.test(resid, lag = 24, type = "Ljung-Box")
#performs the Ljung-Box test on the residuals with a lag of 24 months
print(lb)

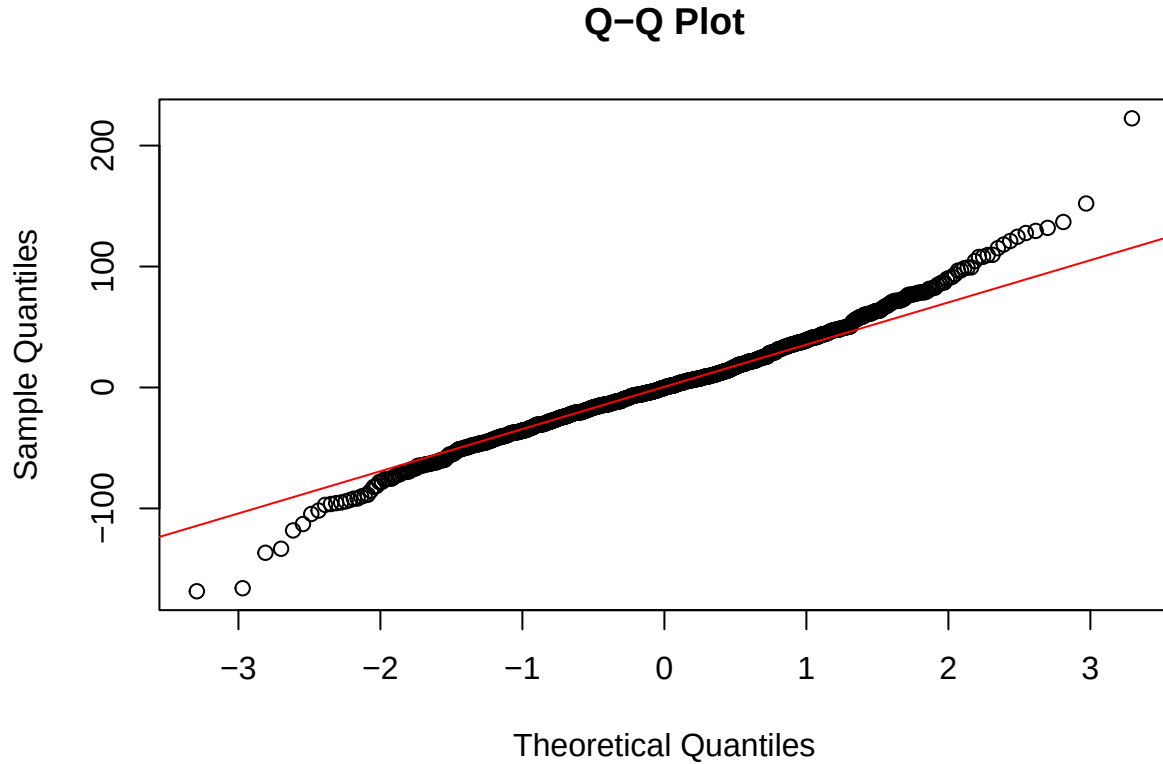
##
## Box-Ljung test
##
## data: resid
## X-squared = 61.459, df = 24, p-value = 3.964e-05

shapiro <- shapiro.test(resid)
#performs the Shapiro-Wilk test on the residuals to
#assess whether they are normally distributed
print(shapiro)

##
## Shapiro-Wilk normality test
##
## data: resid
## W = 0.98318, p-value = 2.206e-09

qqnorm(resid, main = "Q-Q Plot")
#creates a Q-Q plot of the residuals
qqline(resid, col = "red")

```



Conclusion

The present study analyzed monthly sea-level data from 1878 to 1961 using time-series analysis and statistical modeling techniques. A continuous gap-free dataset containing 1,008 monthly observations was selected to ensure reliable analysis without interpolation or missing-value treatment.

The results indicate a clear long-term increasing trend in sea level throughout the study period. Linear Regression, Mann–Kendall, Seasonal Mann–Kendall, and Theil–Sen analyses consistently confirmed a statistically significant positive trend in sea-level rise. STL decomposition further revealed that the observed variability consists of both long-term trend and seasonal fluctuations.

Different visualization techniques, including annual trend plots, anomaly stripes, calendar heatmaps, effectively demonstrated temporal and seasonal variations in sea level. The ARIMA forecasting model showed reasonable short-term predictive performance and successfully captured the historical temporal structure of the dataset.

However, the study also highlights the limitations of statistical forecasting approaches when applied to long-term climate-driven environmental change. Models trained on historical pre 1960 conditions may not fully represent accelerated sea-level rise observed in recent decades due to changing climatic pressures.

Overall, the study demonstrates that time-series analysis provides an effective framework for understanding historical sea-level variability, seasonal behavior, and long-term environmental trends.